Evaluation of an interactive table with tangible objects: Application with children in a classroom

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Abstract
After having introduced interactive tables as educational tools in schools, we present the *TangiSense* interactive table, which we use as the support for our research. We propose to evaluate the *TangiSense* table together with an application, *Recognition and learning of colors*, directly in a nursery school. The objective is to determine the impact of such an educational platform on the children as well as on the teachers. Our article finishes with the first results of our evaluation as well as the prospects envisaged for our future research.

Keywords
Interactive table, tangible and virtual objects, Child Computer Interaction, evaluation

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Introduction
“Take out your calculator”! We have probably all already heard this sentence at school. And yet, it seems highly likely that it will disappear with time. Nowadays, we no longer find just those simple adding machines in the classrooms; we have already moved on to the era of computers and interactive whiteboards. This new technical equipment has made it possible to provide schools and teachers with new ways of teaching and thus of learning. However, whereas computers are now integrated in schools, research work is already focused on a very different platform for interaction: the interactive table (Cf. the growing interest in new conferences, such as ACM ITS “Interactive Tabletops and Surfaces” Conferences 2009, 2010). Their large interaction surface area, along with their capacity to display virtual objects or detect the presence of tangible objects on their surface, lead researchers to believe that they can be excellent tools

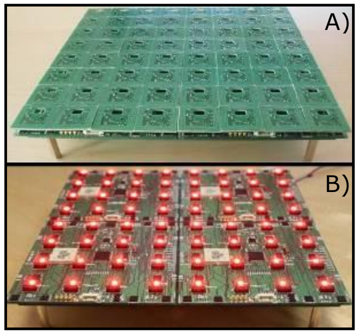

Figure 1. A tile with 8x8 RFID antennas (A), a tile with 1 LED per antenna - prototype v3 (B)

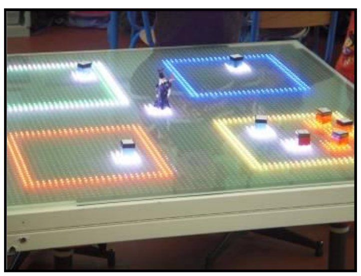

Figure 2. The *TangiSense* interactive table with the "Recognition and learning of colors" application

in the educational world [6]. We propose in this paper to present our *TangiSense* interactive table [3, 4] (which initially comes from the ANR TTT Project, partners: LIG, LAMIH, RFIdées, CEA) and its *Recognition and learning of colors* application. We propose an evaluation of the pair (table, application) in a nursery school. The first results are discussed, as are our ideas on future use of interactive tables in schools for education of children.

TangiSense interactive table

The TangiSense interactive table has the characteristic of allowing interactions with virtual objects (using LEDs or an overhead videoprojection) and also with tangible objects [2]. The interactions with the latter are made possible thanks to the technology used: RFID [1]. The use of RFID makes it possible firstly to detect objects present on the surface of the table (thanks to the different RFID antennas of the table which detects RFID tags), secondly to identify them, thanks to one or more tags stuck underneath the object (a RFID tag being unique), but also to store information directly in these objects or to superimpose them. It is thus possible to work with a set of objects on a table, to store data in these objects (for example their last position) in order to subsequently be able to re-use them on another table at another moment and with your own embedded information (example: the last state of a game of chess). Designed by the RFIdées[1] company, the table is made up of "tiles" of 2.5 cm², each containing 64 antennas (8 x 8), (Figure 1-A), on a surface of 1 m*1 m. Each tile contains a DSP processor, which reads the RFID antennas, the antenna multiplexer, and communication processor. The table contains 25 "tiles" (5 x 5); giving 1600 antennas in total; so 1600 objects could be (theoretically) disposed on the table. Each antenna contains four RGB LEDs (prototype v4) which enable interaction with the users by displaying / lighting some virtual objects, with a very low resolution of course (Figure 1-B). The tiles are associated to a control interface connected to the host computer by an Ethernet link. The table is operational and its hardware / software is now in constant evolution (Figure 2).

[1] http://www.rfidees.fr/

Interactive table for Child Education

Computers are now used more and more in schools. In France, for a few years now for example, certain schools known as "smart schools" have been providing education by using a computer for each child. We think that the interactive tables could be progressively integrated and evaluated in the schools [5], allowing the children to work in a recreational way, either individually or collectively, with the additional possibility of collaboration in the case of collective work. The pupils could work at the same time around the table and could handle objects with the assistance of their teacher (also present at the table). Based on this idea, we proposed an application on our interactive table, *TangiSense*. We asked the teachers how an interactive table and tangible objects could help them in their lessons. It quickly became obvious that the interactive tables with their fun aspect could facilitate the education of children in a good number of areas (error management, group work, real-life situations with objects), to complement a traditional activity using a sheet of paper and glue (*cf*. Figure 3-A). We evaluated the application in a nursery school in order to determine the impact of the use of an interactive table in the school both on the children and on the teachers.

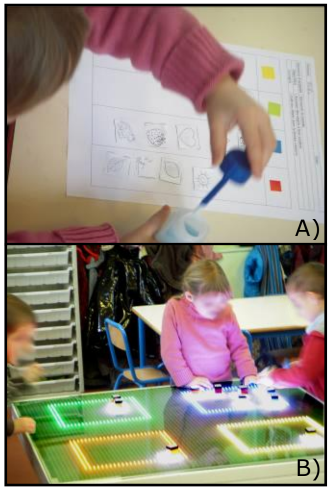

Figure 3. Use of colored sticky labels and paper (A) *versus* the use of the *TangiSense* interactive table (B) in a nursery school

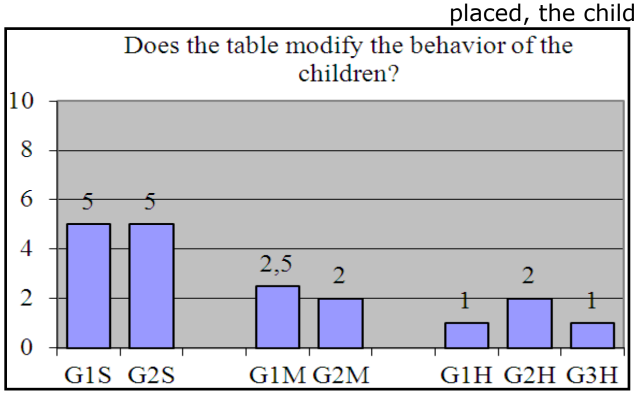

Figure 4. Extract of the first results obtained from the questionnaires filled out by the teacher

Recognition and learning of colors
Application

We propose an application intended to teach the recognition and learning of colors (only red, yellow, blue and green) to children (aged from 2 to 6 according to the level of difficulty). The scenario is based on the French teaching syllabus for nursery schools. We asked a teacher to imagine one or more scenarios using an interactive table and a set of objects without giving any limits or constraints. The teacher proposed a simple application in which the children have to move a set of objects which have "lost their color" into the suitably colored frame (i.e. a "black and white" sun should be placed inside the yellow frame) (cf. Figure 3-B). In order to do this exercise, the child has a set of objects (small cubes which each have an image in black and white). The child then has to determine what the image represents, associate the suitable color to it and place the object within the corresponding colored frame on the interactive table. Once the objects have been placed, the child is invited to check his/her choices using a character representing a magician. The magician then launches the checking procedure, and announces to the child whether he/she has made any errors or has given correct answers (see Figure 2).

Evaluation / First results

The *TangiSense* interactive table was installed in a nursery school and evaluated with 16 children: 5 children in the first nursery class (3 years old approximately) divided into 2 sub-groups called G1S (n=2 children) and G2S (n=3); 5 children in the middle nursery class (4 years old approximately) divided into 2 sub-groups called G1M (n=2) and G2M (n=3); 6 children in the final nursery class (5 years old approximately) divided into 3 sub-groups called G1H (n=2), G2H (n=2), G3H (n=2) in order to estimate the overall impact of its use in school. The teacher explained the work to be carried out to the children: to replace 4 objects within the frame of the suitable color (or in the suitable column on paper). However, the teacher could decide to increase the number of objects according to the aptitudes of the children. The number of objects could then vary to 8 objects per child, integrating more important levels of difficulty. Each group of children did the activity both on paper and on table but not necessary in this order. Indeed, the passages were crossed, when one group was on the table the other one was on paper and *vice-versa*. During the first observations, the children were initially overawed by the interactive table but they were quickly familiarized with it and understood the principle of interaction with tangible and virtual objects well, along with the functions associated to the objects (Figure 3-B). After each session (7 in all), the teacher filled in a questionnaire made up of 10 questions with a value scale from 0 to 10. The questions were aimed at evaluating (a) the perception of the table by the children, (b) the influence of the table on the behavior of the children, (c) the collaboration aspect. The results showed that the children perceived the table rather like a toy, not in the least like a computer, which contributes to the fun aspect added by an interactive table to the work to be done. They also showed, for example, that the table does not modify the children's behavior, nevertheless the professor notified that the small ones (groups G1S and G2S) were initially impressed by the table, for then working naturally (Figure 4). However, the teacher expressed concern that "the exercise, such as it was proposed, did not really allow group work". Nevertheless, our observations clearly show group work involving the children and the teacher, with a lot of interactions and verbal exchanges; each child could take advantage at the same time of the explanations or remarks formulated by the teacher to the others. On the other hand, the teacher gave a very favorable answer to the question "does the table facilitate individual work? " since the exercise was designed so that each child had his/her own objects to replace, and it therefore encouraged the children to work individually.

Conclusion and prospects

In this paper, we have presented the *TangiSense* interactive table, which has the characteristic of allowing interactions with any type of objects (virtual/tangible) and of being able to superimpose the objects and store data inside them thanks to the RFID technology employed. With the help of teachers, we developed an application for nursery school children using the capacities of the table; then we evaluated it in school with children. Following this first evaluation and positive acceptance by both the children and the teachers, we plan to continue our developments and evaluations in schools. The objective would be to provide teachers with a set of tools exploiting the capacities provided by interactive tables such as *TangiSense*. The first results are promising. They encourage us to design other applications and to carry out more research concerning the use of interactive tables as educational tools in schools.

Acknowledgements

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